

Designing Progressive Model Elicitation Tools to Support Complex Cognitive Activities

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Abstract

Externalizations such as sketches and diagrams are effective in helping individuals plan for complex projects and designs. Offloading complex information on a different medium (e.g. paper) frees up cognitive resources for sophisticated thinking. However, there are significant challenges in effectively using externalizations and notations (formalized externalizations). It is difficult to envision formal representations of complex ideas at the beginning and it is also hard to keep track of evolutions within notations. In addition, existing notations may be insufficient or inflexible for our specific purposes. We call instances of such thinking in which humans have to work through complex information and action space as instances of "Progressive Model Elicitation" (PME). We explore ways to support PME processes by presenting a set of design principles and Schematica, a prototype that implements those design principles to support PME. We present illustrative examples of Schematica use to support the process of PME.

CCS Concepts

• **Human-centered computing** → **User models; User models.**

Keywords

Externalization, Notation, Diagramming, Mental Model

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1 Introduction

When a person wants to address a substantial new complex project or design (e.g., the design of a new piece of software or circuit, the set up of a new interdepartmental project for a government, the construction of a new building or even the instructional design of a new course), it is not uncommon to rely on externalizations such as sketches, diagrams, and lists. This is a natural inclination for many of us, and is indeed justified by the power of externalization: by offloading information in the environment (e.g., on paper), that we can retrieve back easily, we might be able to release cognitive resources (e.g., short term memory) and therefore achieve more sophisticated thinking or be able to address topics or systems with complexities out of our capacity [8]. In the limit, if the representations that we create are good enough, they might become formal notations that enable communication and types of thinking that would simply not be accessible otherwise. For example, over the 1990's the software community developed formal visual languages (e.g., UML) to help design and communicate about software behavior and architecture [3]. Another salient example is Richard Feynman's development of a notation for quantum electrodynamics, which some hold responsible for many of Physics' advances in the second half of the twentieth century [15]. As Blackwell and Green argue, good notations and the environments to interact with them can support important cognitive activities [2]. A good notation might make, for example, an impossible or not desirable state of the system (e.g., a contradiction) very obviously visible or, alternatively, simply impossible to create.

However, despite the large potential benefit, there are several fundamental challenges to the effective use of notations and externalizations. The first is that, at the beginning of such a thinking process it is very difficult to know which notation or formalism is appropriate or efficient for our particular problem, or even if it exists. In fact, at the beginning a strict formalism or the requirements of a certain notation for detail might be counterproductive and off-putting¹. A second problem is that a single specific formalism might not be sufficient for our purpose, requiring a mix of several formalisms, an ad hoc formalism or a certain degree of ambiguity

¹a similar argument about the activity of sketching in design is made by Buxton [4]. The ambiguity of sketches is a feature.

while we are figuring out exactly how to externalize information for benefit. A third challenge is that, as our ad-hoc notation evolves, it becomes difficult to adapt the existing body of work to the new evolved constraints and representational conventions. This creates additional work that might, again, detract from the natural flow of the main activity. Moreover, it is often difficult to decide, on the spot, whether it is worth it to revise all the previous material to conform to the evolved state of representation. A fourth and final challenge is that recording and occasionally tracking back through the evolution of the notations can be very difficult due to the continuous and progressive evolution of both the formalization and of the content.

1.1 Progressive Model Elicitation (PME)

We believe that a significant proportion of the instances in which humans have to think through a complex information and action space, such as the architectural, design, or educational examples at the beginning of the introduction, have the characteristics and challenges described in the previous paragraph. We call these instances of *Progressive Model Elicitation* because of their evolving nature and because ultimately, through the externalization and thinking processes, what is being built is a useful model or collection of models of the problem, the actions or the elements involved.

We also argue that existing tools are generally not supportive of the process of progressive model elicitation. For example, even highly-developed generic diagramming or whiteboarding tools such as MS Visio² or Miro³, despite their flexibility and options, are lacking in support of formalizations (e.g., providing representations that are computable or verifiable) or unable to record changes in its use. On the other end, specialized modeling tools adapted to specific domains (e.g., CAD software, financial planning tools) lack the ability to support the early stages of the process, where the thinker is still figuring out even which types of elements are worth externalizing.

Although the academic literature on tools for modeling and visual externalization is too large to review here, we observe that PME is also not covered or considered. For example, a large body of research focuses on the challenges of externalizing specific applications (e.g., programming languages [13], problem representation [27], learning programming and algorithms [10], and argument analysis and creation [1, 5, 6, 22, 25]), which tend to already have pre-established formalisms. Another part of the literature is focused on the interaction challenges of mind mapping (at the bottom level of formalism—see [7, 14]) and sketching (e.g., [19]), and a growing area concerns the integration or infusion of Generative AI in the composition of text diagrams and other representations (e.g., [20, 21, 23, 25]).

2 An Exploration on How to Support PME

In this late breaking work article we explore how to better support PME processes. We collated a set of design principles for interactive tools that could support modeling, and thus thinking, in a more flexible way. The design principles are partly derived logical consequence of supporting the PME process as we have defined and

identified it (see Section 1.1), while also incorporating knowledge from existing theories of cognition and evidence of how people use externalization. We used these principles to design a prototype tool, Schematica, that implements a few features to support PME. We present Schematica through the scenario of a fictitious city official charged with starting a sustainability plan for the Open Geofiction city of Saindzaul⁴. We finalize with a description of additional designed features not yet in the system.

2.1 Formalization and Design Principles for PME

A core conceptual element of our approach to support PME is that externalizations exist in a continuum of formalization. For example, at the “non-formalized” end of the continuum, a sketch on the proverbial back of the envelope with a few words and a couple of lines can be a suitable starting point for ideation. This extreme example has hardly any formality: the words can represent ideas, actions, concepts or almost anything, and the lines might be equally ambiguous, representing any relationship between the elements. It is, however, very quick to make and can be extremely efficient to support memory (e.g., if an idea came up while on-the-go) and communicate with others during an informal conversation).

An example of a next step in formalization is the transformation of that initial sketch into a *mind map*. In a mind map there are generally just two types of elements, ideas, and connections, but it does already impose some formality because an element should only contain one idea, and connectors are generally only allowed to connect a pair of elements.

One can imagine how mind maps can, in turn, be transformed into more sophisticated variants where its ideas can instead be classified into different classes of objects, where connections can be uni- or bi-directional, of different types themselves, and have a name or a title. Eventually we reach the other end of formality, usually purpose specific, in the form of notations that might have additional power, such as supporting certain types of computations. For example, the Bayesian Network formalism [17] includes random variables (nodes) that are connected to each other through edges (uni-directional links), where each node has a table of probabilities that connects its possible values dependent on the incoming links (these could be implemented as attributes), and with some overall rules, such as the graph formed by connecting variables to each other through links has to be acyclic. Another example is the adoption of an argumentation ontology to describe and calculate support for political arguments [5, 6]. Therefore we adopt **DP1: Support Flexible and non-committal traversal of the continuum of formality**. Because both the quick and ambiguous sketches at one end, and the inconsistency-detection and inference-capable formalisms at the other are desirable, a system and process for PME should support as much of the continuum as possible and remove the barriers to transition between levels of formality in both directions. A corollary is that, although a system might encourage or help thinkers to increase in formality as the process progresses, it should not prevent the interface from working or even remaining looking pleasant just because of a lack of further formalisms. This would be counterproductive and discourage use at early stages. DP1

²<https://www.microsoft.com/en-gb/microsoft-365/visio/flowchart-software>

³<https://miro.com>

⁴<https://wiki.opengeofiction.net/index.php/Saindzaul>

aligns with the Cognitive Dimensions of Notation framework's call to avoid *premature commitment*.

Many software systems that support formalization do it only for a single formalism and assume that a single repository/main domain is being worked on at a given time. This is only natural, since it is difficult and expensive to make “everything tools” that can cover all the representational needs of a person or team. However, it seems also clear that, in any serious design or thinking process, multiple representations and models are required. This is because a single representation is rarely able to convey all the useful information or support all the useful actions of the process. Some previous research has also shown that people naturally create multiple representations and that the transformation of a representation into another is a rich process that might be key for successful modeling and thinking [1]. For this reason we adopt **DP2: support multiple representations and annotation**. Modelers should be allowed to create a new version of the information at any given time without detriment of the other representations. The ability to annotate or support a *secondary notation* [2] is also crucial so that information can be committed to the representation without feeling that it is being made murky, diluted, or harder to access. DP1 and DP2 are supported by evidence that, when free to create notations, sketchers and designers tend to mix and invent representations [24, 26]. More generally, the importance of a variety of representations has been highlighted much earlier (e.g., within the theory of Distributed Cognition [11, 12]).

At the same time, multiple representations are also unlikely to be completely unrelated or unconnected. It is likely that some elements that appear in one diagram also appear in another. Moreover, processes of transformation between a representation and another are cognitively rich and typically not one-to-one (i.e., an object in Representation A might convert into several objects in Representation B—an expansion—or vice versa—a summarization). For this reason we posit **DP3: facilitate transition and connections between representations**. Tools that support sophisticated ways to derive elements in one representation from another representation and also keep track of the relationships and provenance of objects in any of the representations may increase the sophistication of cognitive processing that is possible and reduce the cognitive load of having to remember how objects were created. This principle is supported by observations of people in semi-controlled environments, who tend to create multiple representations and often struggle to maintain them synced [1].

Beyond DP3, it seems reasonable that tracking the creation and manipulation of the formalism itself is also important. This is because the process of creating a formalism is unlikely to be straightforward. Modelers will try formal constraints that they might have to roll back, or new kinds of objects or links that are then merged or split according to the formal needs of the problem at hand. To support this, systems will have to **DP4: capture and record status of the formalization as well as support selective undoing or redoing of application of the formal constraints**. Together with DP3, DP4 makes an argument for considering *interaction provenance* [18] from scratch.

Finally, the process of developing the right set of formalizations alongside the understanding of the subject matter is valuable in itself. Therefore the formalism might be applicable to other projects

or by other people. Moreover, being able to reflect and comment on the formalization and process aspects of the project itself is likely to be valuable for the practitioner. For this reason, we encourage **DP5: support reflection, systematization and sharing of the process**. We posit DP4 and DP5 based on our own experience of implementing and using informal notations (and trying to recover their meaning after several months of hiatus). However, these also align with templating and commenting features that we increasingly observe in commercial tools (e.g., Miro).

In the following Section we describe some of the ways in which we have been able to apply design principles 1–5 in the design of the features of a prototype experimental tool that is still in development: Schematica. The Section afterwards describes examples of advanced features that have not yet been implemented.

3 Schematica for Saindzaul Climate Change Planning

We introduce Nina, a climate scientist who has been hired to create an Office of Climate Change management to tackle the growing climate crisis for the fictional coastal city of Saindzaul.

Brainstorming. Nina is preparing for her role and would like to map out some of her preliminary thoughts. She opens Schematica and creates a new map.

She starts by writing down some ideas on how to create this new office. She writes down items as nodes (by double clicking anywhere on the canvas) including “waste management” and “carbon taxing” along with “budget” and some more. We refer to this map as the *Notes Map* (Figure 1 A). Looking at her notes, she realizes that what she has written is heterogeneous and difficult to interpret. She would like to add some more structure to her ideas.

She creates two categories for “Logistics” and “Operations” (by clicking on the text input at the top of the node) and assigns the nodes the categories to represent the broad considerations that she will have to make while creating her department (DP1)(Figure 1 B). She assigns “budget” and “staffing” to logistics since these things would need to be decided for the new climate office, while the rest are more fittingly operations since they are already being managed by existing agencies. She finds this categorization to be clarifying for her to see that logistics and operations require different considerations. She moves the logistics nodes to the left and wants to further expand the scope of the operations.

Structuring thoughts. Nina knows that she will have to work with the municipal and provincial governments to set up and operate the office. Several of the roles that will fall under the new climate change office are currently being handled by different government agencies. To account for this, Nina creates two new nodes of type “Municipal Gov” and “Provincial Gov” and places them above the other nodes to denote their current supervisory roles (Figure 1 C). She finds that arranging the operations under the two different government structures along with moving logistics to the side is useful to understand the agencies that she will need to work with (DP2). She also creates directional links (>>>) between government bodies (by clicking the + on the top left of the nodes) and various operations. She creates a new type for the linkages as “in charge” (DP2, DP3) to denote the various operations that fall under the

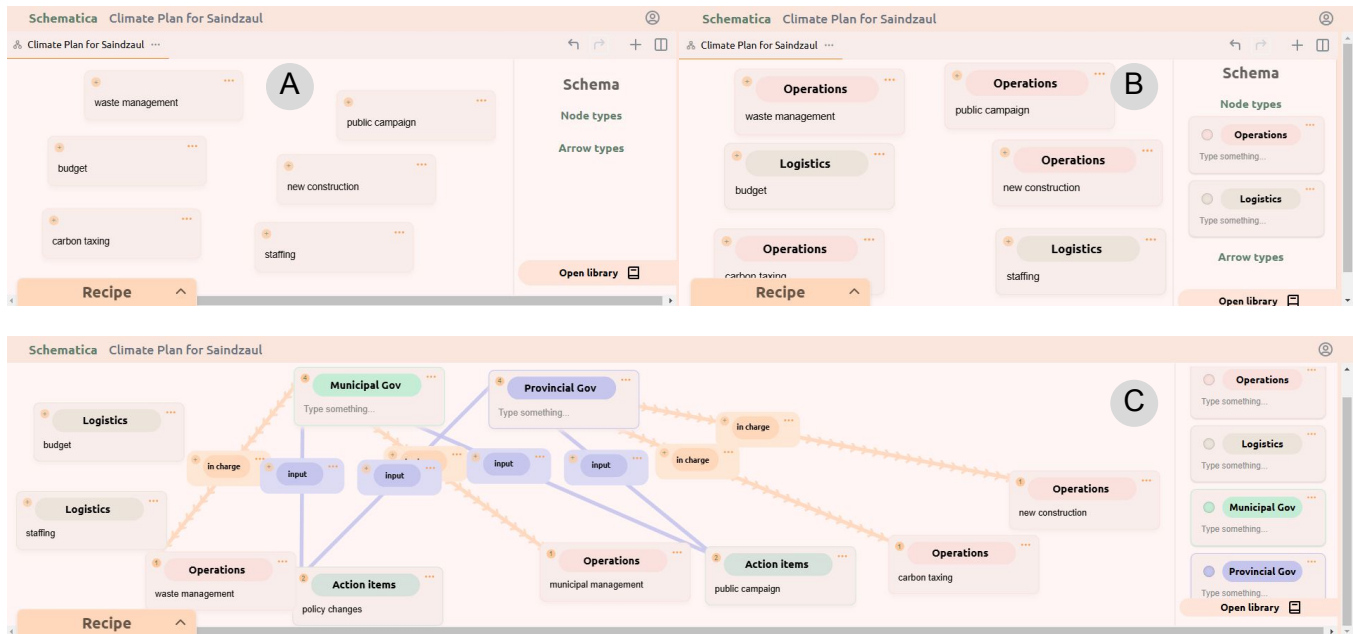


Figure 1: Schematica Interface. ‘Notes Map’ (A) and progressive structured analysis: adding categories (B), and links (C).

responsibilities of the different governments. She realizes that in addition to being in charge of certain operations, the governments will be expected to provide input for various operations as they are moved into the new climate change office. She creates links between the government nodes and the various operations where their input is required and creates a new type of linkage called “input”. Nina finds that adding formality to this map is useful for organizing her ideas. Nina realizes that “policy changes” and “public campaign” are action items rather than operations since these still need to be implemented. She clicks on the text input for the two respective nodes and changes the type to “Action items” (DP1, DP4).

Nina is able to get a clearer idea of the climate change office she plans to develop and lead. She wants to add more specific information about the division of responsibilities across different aspects of the office. She adds an attribute (=responsibility=) (DP2, DP4) to the logistics node on the schema pane (left - Figure 2 A), to denote the level of responsibilities that her office will have regarding logistics. She also adds attributes for the estimated cost and timeline (=estimated cost= and =timeline=) for the action items. At this point, Nina has a clearer idea of the distribution of responsibilities, and budgets across different departments. We refer to this view as the *Responsibilities Map* (Figure 2 A). Having a more formal and structured representation of her ideas is proving to be useful for Nina to make plans for this complex office. After mapping out the different operations that the climate change office will be taking over, Nina wants to further map out the operations and the planning. She wants to get a sense of the different priorities and timelines that she would need to follow. She feels that the current map has become too cluttered and it is a rather different perspective she wants to analyze (DP2). She, therefore, creates a new map (by clicking on the + button) and calls it *Timeline and Priority* (Figure 2 C).

Alternative views. The schema that Nina has created so far in her Responsibilities Map has been saved on the right hand side panel of Schematica including nodes and link types together with their attributes. Hence, Nina can easily import this previous schema into the Timeline and Priority (DP3, DP4). She, then, drags the operations and planning nodes to the new map (DP3). She modifies the node types in the schema for operations and planning to include the =timeline= (denoting the estimated time by which the items are expected to be in place) (DP3, DP4). She arranges the nodes horizontally in terms of the timeline for when she hopes to accomplish the transfer of operations and completion of planning and also arranges them vertically in terms of priority (Figure 2 C). Arranging her thoughts with escalating levels of formality has made the planning for this new climate office more intuitive and insightful for Nina. She wants to save her progress and also wants to share it with other members in her team. She makes notes of her thought process in the “Recipe” panel to help others understand her procedure and to be able to reproduce it (Figure 2 B) (DP5).

4 Future work: Additional features

In the previous section we discussed currently implemented features of Schematica that constitute the basis for a progressive model elicitation tool. Here we present several additional features that may complement our tool to further support better reasoning about a complex problem.

Supporting Transitions and Intermediate Representations. Schematica allows for a multitude of representations as shown by the Responsibilities and Timeline and Priority maps discussed in our example (DP2). We stress that for a system to support a user to navigate the continuum of formality each representation should not be seen as the final product, but rather as a step towards such

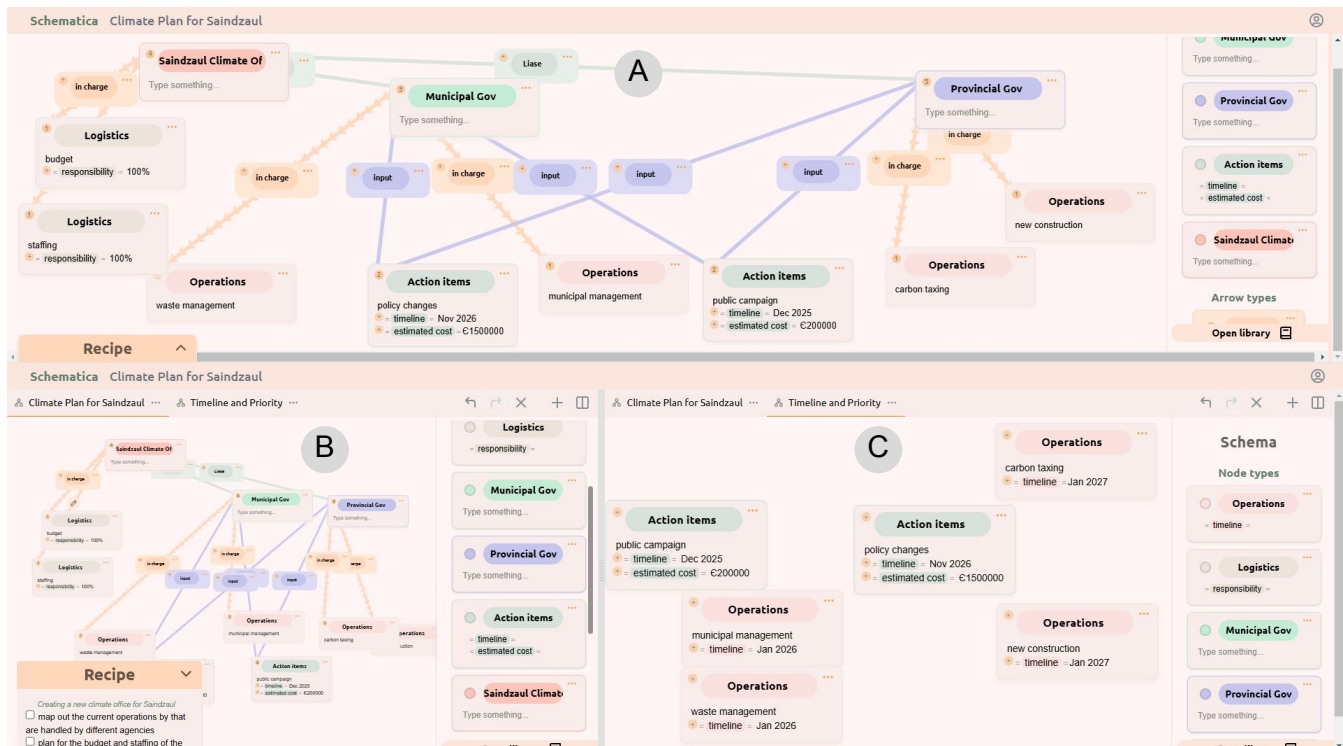


Figure 2: Schematica Interface. ‘Responsibilities Map’ with attributes (A), and recipes (B). ‘Timeline and Priority’ map (C).

a product. In our tool, we consider intermediate drafts, obtained through transformations of components to other components, as first-class representations, able to convey information, as long as they are self sufficient but conceptually or visually disjoint (following [1]). For this reason, our design of Schematica will allow to access intermediate representations and use them to provide different views or analysis of information (DP1, DP3, DP4). For the Saindzaul climate plan, assume there are further important aspects, for example all items in the climate plan must be planned considering potential effects on the public opinion. Nina may wish to analyse this aspect separately from the previous maps, yet starting from her same initial set of notes. In Figure 3 we demonstrate some extract of the two transformations, starting from the center (B), Nina could construct the Responsibilities map first (C) but then go back to the original notes (B) to build a pros and cons representation of the public opinion on goals proposed by the climate plan (A). Our tool will seamlessly permit navigation between saved maps, and intermediate representations through a historical navigation pane, through different levels of formality. Our design is intended to allow a user to access each step in the creation and re-elaboration of representations to achieve a desired level of formality. This is helpful to understand how a representation has been reached and what steps have inspired different threads of reasoning and idea creation reducing the burden on the user to remember how elements were created.

Supporting Intermap Associations. With focus on the individual components, we note that nodes in different maps may be given

different annotations depending on the context and meaning of a specific representation (DP2). For example the node “waste management” (Figure 3 B) constitutes an operation overseen by one of the government of Saindzaul in the Responsibilities Map (Figure 3 C). On the other hand, it is one of the goals in the climate plan program that must be considered for public scrutiny (Figure 3 A). Our design is so that there exist intermap association relationships between items used in different maps, such that one can easily inspect where and how a thought has been elaborated (DP3). Assuming that only one-to-one associations between components exist between representations has severe limitations. Previous research showed that transformation between components of a representations may have different levels of granularity, where many-to-one or one-to-many are as common as one-to-one associations [1] and help a user progressively build an understanding of the problem at hand. All these views should be afforded by the system (DP2-DP3), able to provide flexibility in turning an element of a starting representation into many elements of a new representation and vice versa. Additionally, it should allow for expansion or summarization of text or properties of a node into a new node, without losing association between the nodes. For example, waste management is split into many functionalities in Figure 3 F, while Figure 3 E shows a node expansion in the textual description of such processes.

Supporting Higher Levels of Formality. A key feature of a progressive model elicitation tool is the ability to support higher levels of formality (DP1). We envisage Schematica to provide a library of commonly available formalisms including, for example, UML,

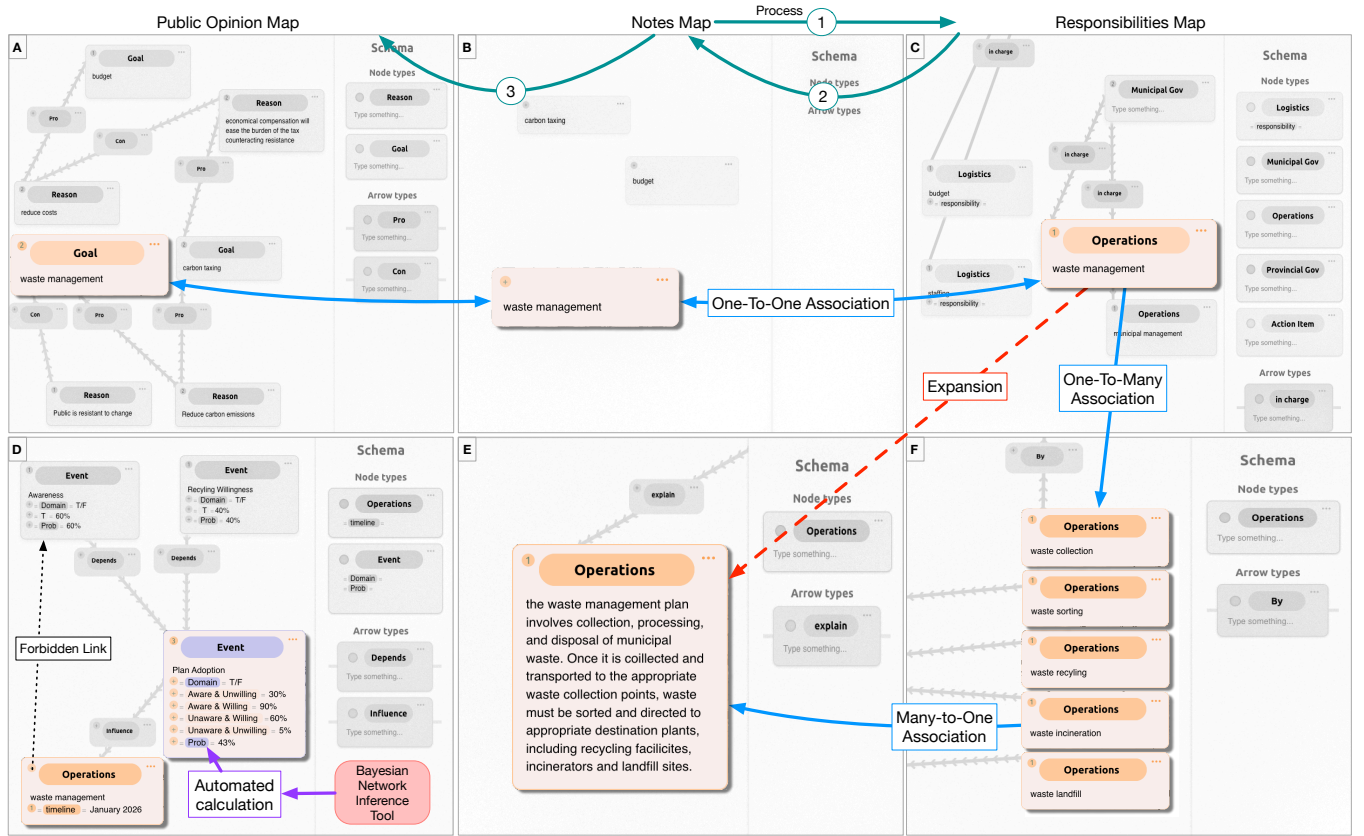


Figure 3: Future Additional Features. Supporting representation transitions (A,B,C), intermap association (B,C,E,F) and higher levels of formality with automated tools (D).

and Bayesian Networks, as discussed in Section 2. By complying with existing notations for part of the problem under consideration, these formalisms can in turn be useful to aid complex cognitive tasks. Formal representations may complement the current reasoning process with ways to clearly structure thoughts, enable specific computations or identify issues and errors. Some notations may even facilitate the use of various automated reasoning tools, for example inferential reasoning methods in Bayesian Networks [17]. Progressing in our example, we may assume that to plan for waste management operations, the new government must take into consideration factors that may influence the likelihood of adoption of such a plan, including for example public waste awareness, willingness of local population to engage in recycling, etc. This map can be represented as a Bayesian Network and the resulting plan adoption likelihood may then further influence the timeline (Figure 2 C). After nodes and connections are created among various factors, appropriate values can be filled for conditional probability tables. The tool would permit the automatic calculation of posterior probabilities (Figure 3 D). Nina would then be provided with an estimated likelihood of adoption of the waste management plan according to the population awareness and willingness to engage in recycling. The process of construction of these maps would be recorded in the recipe, but in addition, we aim to be able to make our recipes

interactive, such that a user is able to automatically run some of the formalizing steps, which would become more easily shareable and repeatable by other users (DP5). Finally, given that our proposed PME tool designed for flexibility (DP1), a variety of such reasoning tool can be included, for example one may determine the acceptability of the set of goals in the agenda according to public opinion (Figure 3 A) according to automatic argumentation techniques [22]. The ability to adhere to a specific formalism, and to arrive at this level progressively, means that one must also adopt a progressive introduction of constraints between components of each formalism. For example, a Bayesian network is a directed acyclic graph, where nodes have to be unidirectional, and hence, the tool must check for cycles and prompt the user to correct such potential issues. Future work may also consider how the tool can suggest schemes from the library similar to the current representation through different metrics [16].

5 Discussion, Limitations and Conclusion

Complex projects that require complex thinking are likely to benefit from tools that support advanced externalization. In this paper we propose that one promising avenue to support this kind of work is to create designs that consider thinking activities from the lens of Progressive Model Elicitation (PME). PME removes the

assumption that formal languages or notations are complete and usable at the beginning of the thinking process, and instead suggest that notations and formalizations are progressively constructed in parallel to the thinking about the subject matter itself.

In this late breaking work submission we present the design principles that we have derived, and the instantiation of a series of features on a prototype tool called Schematica, that exemplify how PME can be supported. We believe that this approach opens several perspectives and opportunities to improve on cognitive support tools for complex activities, including wicked problems [9].

Although Schematica provides a good starting point for the exploration of the value of supporting PME, the prototype tool is limited in several important ways. For example, although boxes/circles and arrows/links are dominant as visuals and are commonly used and understood by people who sketch and externalize ideas (e.g., [24, 26]), many other variants and visual appearances are possible. For presentation purposes, we used Nina's climate planning task as an example to demonstrate how a PME-enabling tool is beneficial in supporting a complex reasoning process composed by several evolving representations including notes, concept maps, timelines, argument maps, and Bayesian networks. Due to its flexibility, we envisage many other use cases for a PME tool including brainstorming ideas for research projects or papers, analyzing factors affecting a specific problem, such as a political or legal issue, forecasting, and mapping information flows between organizations. While our main case study starts from representations with no formality, a user can start at any level, for example with a prescribed formal representation, to then create less formal representations for better explaining concepts to their audience. Additional use cases or limitations may emerge as our PME approach undergoes further development.

Many more features are possible and future designs might also offer better alternative features, including GenAI-infused features that can simplify the more tedious consequences of having to keep track and interpret the evolution of notations in parallel with the modeling itself. Additionally, we still need to validate the assumption that PME is indeed as common as our personal experiences suggest, and to assess the value of supporting it in real-life processes.

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